Part I

Chapter 2 Exercises

Consider the least squares loss expression

$$L[\phi] = \sum_{i \in I} (f[x_i, \phi] - y_i)^2$$

for the linear regression of two variables,

$$f[x_i, \phi] = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix} = X\phi$$

Exercise 1. Calculate the partial derivative of f with respect to each ϕ_i, known as the gradient.

Solution 1. Observe the gradient of f is

$$\begin{aligned}
\nabla f[x_i, \phi] &= \left\langle \frac{\partial f}{\partial \phi_0}, \frac{\partial f}{\partial \phi_1} \right\rangle \\
&= \left\langle \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle \\
&= \langle 1, x_i \rangle
\end{aligned}$$

Exercise 2. Find the minima of $L[\phi]$.

Solution 2. The minima occur at the critical points of L, namely where $\nabla_\phi L = 0$. Since nabla is a linear operator, we can pass it over the sums and also invoke the chain rule, i.e.,

$$\begin{aligned}
\nabla_\phi L[\phi] &= \nabla_\phi \left(\sum_{i \in I} (f[x_i, \phi] - y_i)^2 \right) \\
&= \sum_{i \in I} \nabla_\phi (f[x_i, \phi] - y_i)^2 \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \nabla_\phi (f[x_i, \phi] - y_i)) \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot (\nabla_\phi f[x_i, \phi] - \nabla y_i)) \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \langle 1, x_i \rangle)
\end{aligned}$$

Thus

$$\begin{aligned}
\nabla_\phi L[\phi] &= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \langle 1, x_i \rangle) \\
&= \sum_{i \in I} \langle 2(f[x_i, \phi] - y_i), 2(f[x_i, \phi] - y_i) x_i \rangle \\
&= 2 \sum_{i \in I} \langle f[x_i, \phi] - y_i, (f[x_i, \phi] - y_i) x_i \rangle
\end{aligned}$$

Next, to find the critical points, $\nabla_\phi L[\phi] = \widehat{0}$ occurs for each i,

$$2 \sum_{i \in I} \langle f[x_i, \phi] - y_i, (f[x_i, \phi] - y_i) x_i \rangle = \langle 0, 0 \rangle$$

for the first entry, we have

$$\begin{aligned}
\sum_{i\in I}\left(f\left[x_i,\phi\right]-y_i\right)&=0\\
\sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-y_i\right)&=0\\
\sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]\right)-\sum_{i\in I}y_i&=0
\end{aligned}$$

and solving for ϕ_0 yields

$$\begin{aligned}
\sum_{i\in I}\left(\phi_0+x_i\phi_1\right)-\sum_{i\in I}y_i&=0\\
\sum_{i\in I}\phi_0+\sum_{i\in I}x_i\phi_1-\sum_{i\in I}y_i&=0\\
\phi_0\left|I\right|+\phi_1\sum_{i\in I}x_i-\sum_{i\in I}y_i&=0\\
\frac{\sum_{i\in I}y_i-\phi_1\sum_{i\in I}x_i}{\left|I\right|}&=\phi_0\\
\overline{y}-\phi_1\overline{x}&=\phi_0
\end{aligned}$$

For the second entry, we have

$$\begin{aligned}
\sum_{i\in I}\left(f\left[x_i,\phi\right]-y_i\right)x_i&=0\\
\sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-y_i\right)x_i&=0\\
\sum_{i\in I}\left(\left[\begin{array}{cc}x_i & x_i^2\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-x_iy_i\right)&=0
\end{aligned}$$

and solving for $\phi_1$yields

$$\begin{aligned}
\sum_{i\in I}\left(x_i\phi_0+x_i^2\phi_1-x_iy_i\right)&=0\\
\sum_{i\in I}x_i\phi_0+\sum_{i\in I}x_i^2\phi_1-\sum_{i\in I}x_iy_i&=0\\
\phi_0\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0
\end{aligned}$$

substituting ϕ_0 yields with

$$\begin{aligned}
\left(\overline{y}-\phi_1\overline{x}\right)\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0\\
\overline{y}\sum_{i\in I}x_i-\phi_1\overline{x}\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0\\
\frac{\overline{y}\sum_{i\in I}x_i-\sum_{i\in I}x_iy_i}{\left(\overline{x}\sum_{i\in I}x_i-\sum_{i\in I}x_i^2\right)}&=\phi_1
\end{aligned}$$

Rewrite the numerator as

$$\begin{aligned}
\overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i &= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + 0 \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \left(\overline{y}\sum_{i\in I} x_i - \overline{y}\sum_{i\in I} x_i\right) \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \frac{\sum_{i\in I} y_i}{|I|}\sum_{i\in I} x_i - |I|\,\overline{y}\frac{\sum_{i\in I} x_i}{|I|} \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \overline{x}\sum_{i\in I} y_i - |I|\,\overline{y}\overline{x} \\
&= \sum_{i\in I} \overline{y}x_i - \sum_{i\in I} x_i y_i + \sum_{i\in I} \overline{x}y_i - \sum_{i\in I} \overline{y}\overline{x} \\
&= \sum_{i\in I} \left(\overline{y}x_i - x_i y_i + \overline{x}y_i - \overline{y}\overline{x}\right) \\
&= \sum_{i\in I} \left(x_i\left(\overline{y} - y_i\right) - \overline{x}\left(\overline{y} - y_i\right)\right) \\
&= -\sum_{i\in I} \left(x_i - \overline{x}\right)\left(y_i - \overline{y}\right)
\end{aligned}$$

Rewrite the denominator as

$$\begin{aligned}
\overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2 &= 2\overline{x}\sum_{i\in I} x_i - \overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2 \\
&= \sum_{i\in I} 2\overline{x}x_i - \sum_{i\in I} \overline{x}^2 - \sum_{i\in I} x_i^2 \\
&= -\sum_{i\in I} \left(x_i^2 - 2\overline{x}x_i + \overline{x}^2\right) \\
&= -\sum_{i\in I} \left(x_i - \overline{x}\right)^2
\end{aligned}$$

Finally,

$$\phi_1 = \frac{\overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i}{\overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2} = \frac{\sum_{i\in I}\left(x_i - \overline{x}\right)\left(y_i - \overline{y}\right)}{\sum_{i\in I}\left(x_i - \overline{x}\right)^2}$$

Hence,

$$\widehat{\phi} = \begin{bmatrix} \overline{y} - \phi_1\overline{x} \\ \frac{\sum_{i\in I}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i\in I}(x_i - \overline{x})^2} \end{bmatrix}$$

Exercise 3. Reformulate the linear regression as a generative model, i.e.,

$$x = g\left[y, \phi\right] = \begin{bmatrix} 1 & y \end{bmatrix}\begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix}$$

(a) Find the new loss function and the inference expression, $y = g^{-1}\left[x, \phi\right]$.

(b) Verify if the model will make the same predictions as the discriminate version. Use code if you need to.

Solution 3. We'll solve in parts.

(a) Since least squares loss is defined using the difference between the input and the target variables, the least squares loss function for the generative model $x = g\left[y, \phi\right]$ can be expressed as

$$L\left[\phi\right] = \sum_{i\in I}\left(x_i - g\left[y_i, \phi\right]\right)^2$$

The inference expression can be solved for in terms of y

$$X = Y\phi \implies Y = X\phi^T\left(\phi\phi^T\right)^{-1}$$

But we're lazy, so we'll just expand the inner product and solve for y,

$$\begin{aligned}
x &= \begin{bmatrix} 1 & y \end{bmatrix}\begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix} = \phi_0 + y\phi_1 \\
\implies y &= \frac{x - \phi_0}{\phi_1} \\
&= \frac{x}{\phi_1} - \frac{\phi_0}{\phi_1} \\
&= \begin{bmatrix} 1 & x \end{bmatrix}\begin{bmatrix} -\frac{\phi_0}{\phi_1} \\ \frac{1}{\phi_1} \end{bmatrix}
\end{aligned}$$

where $\phi_1 \neq 0$

(b) Fitting $y \sim f[\phi, x]$ via linear regression wouldn't be the same as fitting $x \sim g[\theta, x]$ unless their parameters satisfy

$$\widehat{\phi}_1 = \frac{1}{\widehat{\theta}_1}$$

or, specifically,

$$\frac{Cov(x, y)}{Var(y)} = \frac{1}{\frac{Cov(x,y)}{Var(x)}}$$

This happens when the modeled data is strictly linear.